

Deep Exemplar-Based Video Colorization with Global Temperature Optimization, Edge-Preserving Chromatic Smoothing, and Bidirectional Temporal Fusion

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Abstract

Video colorization aims to reconstruct plausible chromatic information for grayscale video sequences while maintaining semantic fidelity, spatial coherence, and temporal consistency. Compared with single-image colorization, video colorization is considerably more challenging because independently generated frame colors may fluctuate over time, causing flickering, color fading, boundary bleeding, and abrupt chromatic transitions. Exemplar-based video colorization partially alleviates color ambiguity by transferring chromatic information from selected color reference frames; however, its performance strongly depends on the quality of semantic correspondence, the selection of color-estimation parameters, spatial smoothing behavior, and the strategy used to propagate colors between reference keyframes. Recent approaches have explored transformer-based color representations, generative priors, semantic correspondence, bidirectional temporal feature propagation, diffusion models, and multimodal guidance, but reliable temporal stability remains difficult under motion, appearance changes, imperfect exemplar matching, and long temporal intervals. This paper proposes a deep exemplar-based video colorization framework that improves a recurrent reference-guided colorization architecture through three complementary mechanisms. First, a global temperature coefficient is estimated before full-video inference by minimizing the average L2 chromatic reconstruction discrepancy over representative reference-keyframe pairs. Second, the parameters of a weighted least-squares filtering stage are optimized to spread chromatic information smoothly while preserving meaningful object boundaries. Third, a bidirectional temporal fusion strategy generates two color estimates for every frame between successive keyframes and combines them according to normalized temporal distance, thereby eliminating the abrupt reference switching that produces visible flicker in conventional keyframe-centered processing. The resulting framework is designed to improve color fidelity, reduce localized color contamination, and increase temporal smoothness without requiring fundamental retraining of the underlying exemplar-based network. A comprehensive evaluation protocol is formulated using image-quality, perceptual, color-distribution, and temporal-consistency criteria on established video-colorization benchmarks.

Keywords: video colorization; exemplar-based colorization; deep neural network; semantic correspondence; temporal consistency; weighted least squares; parameter optimization; bidirectional temporal fusion; keyframe color propagation.

1. Introduction

Colorization is the process of inferring plausible chromatic information from a grayscale image or video. In a luminance-chrominance representation such as CIE Lab, the problem can be interpreted as estimating the missing a and b chrominance components from the observed luminance channel L. This mapping is fundamentally ill-posed because a single grayscale observation may correspond to many physically and semantically reasonable color solutions. A grayscale automobile, garment, building, sky, or natural object can often be assigned several different colors without contradicting its luminance structure. Consequently, an effective colorization system must do more than reconstruct low-level image intensities: it must infer semantic identity, object boundaries, material properties, contextual relations, and plausible color distributions [1].

Deep image-colorization research has advanced significantly through convolutional networks, generative models, transformers, learned color memories, multimodal conditioning, and diffusion-based generation. ColorFormer, for example, combines a transformer encoder with a color-memory decoder and global-local hybrid attention to improve semantic consistency and color richness [2]. UniColor formulates automatic, exemplar-, text-, and stroke-guided colorization within a unified transformer architecture [3]. BigColor learns a generative color prior focused specifically

on synthesizing plausible color conditioned on spatial structure [4], whereas DDColor uses pixel and query-based color decoders to connect multiscale semantics with chromatic predictions and suppress color bleeding [5]. These developments demonstrate that semantic understanding and strong color priors substantially improve still-image colorization.

The extension from image colorization to video colorization introduces an additional and particularly demanding constraint: temporal consistency. A visually plausible color generated for one isolated frame may change unexpectedly in the following frame, even when the physical object itself changes only slightly. Small prediction differences accumulated across time can appear as flickering, intermittent saturation, unstable object colors, color fading, or abrupt transitions. The problem is especially severe when a network processes individual frames independently or when propagated colors gradually drift away from the intended reference. Recent surveys identify spatial-temporal consistency as one of the central unresolved problems in video colorization, and commonly used evaluations therefore combine image-quality measures with dedicated temporal-consistency metrics [13].

Exemplar-based colorization addresses part of the ambiguity by supplying a color reference image from which chromatic information can be transferred to semantically corresponding regions of a grayscale target. In a video setting, selected color keyframes may serve as exemplars for neighboring grayscale frames. This approach is attractive for historical-film restoration, artistic post-production, animation, and other cases in which a human operator can provide or approve representative color frames. However, exemplar-based systems are highly sensitive to correspondence errors. When an object, pose, illumination condition, or scene layout in the target frame differs substantially from the reference, chromatic information may be transferred to an incorrect semantic region.

Recent video-colorization methods have attempted to resolve these limitations through semantic correspondence and explicit temporal modeling. Temporal Consistent Automatic Video Colorization uses a reference image together with the immediately preceding colorized frame to improve both short- and long-range consistency [7]. TCVC propagates deep frame-level features bidirectionally and introduces self-regularization to reduce prediction discrepancies across time [8]. BiSTNet establishes semantic correspondence with exemplar images and combines it with bidirectional temporal feature fusion and semantic boundary guidance [9]. A separate long-term spatiotemporal framework combines convolutional and transformer components with a linkage subnet to improve robustness under irregular motion [10]. These studies confirm that bidirectional information, long-range dependency modeling, semantic priors, and temporal fusion are highly valuable for stable video colorization.

Nevertheless, three practical problems remain insufficiently addressed in many exemplar-based pipelines. First, the behavior of semantic color transfer depends on the sharpness of the correspondence distribution or annealing operation. A fixed temperature-related coefficient can work well for one scene but poorly for another because different videos contain different brightness distributions, chromatic diversity, semantic complexity, and reference-target similarity. A manually selected value is also subjective and difficult to generalize.

Second, raw color estimates frequently require spatial refinement. Strong smoothing can suppress isolated color errors but may also blur object boundaries and spread chromatic information across unrelated regions. Weak smoothing preserves edges but may leave mottled, incomplete, or locally inconsistent color. Therefore, the smoothing strength and boundary-preservation behavior should be determined objectively instead of being selected only by manual inspection.

Third, conventional keyframe-centered processing can introduce a hard transition in the middle of the interval between two keyframes. If the first half of an interval is colorized exclusively from the earlier keyframe and the second half exclusively from the later keyframe, the reference source changes abruptly at the midpoint. Even when both individual colorizations are reasonable, their chromatic distributions may differ enough to create a perceptible blink or discontinuity.

To address these limitations, this paper introduces a three-part enhancement to a deep exemplar-based video-colorization framework. The first contribution is a normalized global temperature coefficient T_{opt} estimated automatically from several representative keyframe-reference pairs. Candidate coefficient values are evaluated by recolorizing selected grayscale keyframes and measuring their discrepancy from corresponding known color frames. The coefficient that minimizes average reconstruction error is selected for subsequent whole-video inference.

The second contribution is joint optimization of two weighted least-squares filtering parameters. One parameter controls the spatial spread of estimated chromatic information, whereas the second controls the degree of edge or boundary preservation. The optimal pair is chosen by minimizing the average discrepancy between filtered colorization outputs and representative ground-truth color keyframes.

The third contribution is bidirectional temporal fusion. Every frame between successive keyframes receives two candidate color estimates: one conditioned on the preceding reference and one conditioned on the following reference. The two estimates are blended using complementary temporal-distance weights. Consequently, the influence of the earlier keyframe decreases progressively while that of the later keyframe increases, avoiding the abrupt midpoint switch inherent in the original scheme.

Recent diffusion-based and multimodal approaches such as OmniFusion, LatentColorization, L-C4, VanGogh, and palette-guided video diffusion demonstrate the broader trend toward stronger priors, richer conditioning, and explicit temporal stabilization [11,12,14-16]. The present study follows a complementary direction: rather than replacing the exemplar backbone with a substantially larger generative architecture, it improves practical reference-guided video colorization through objective parameter estimation, edge-aware refinement, and mathematically simple bidirectional temporal fusion.

The remainder of this paper is organized as follows. Section 2 reviews recent research on deep image colorization, exemplar-guided colorization, and temporally consistent video colorization. Section 3 presents the complete proposed method. Section 4 specifies implementation details, evaluation criteria, and the experimental analysis framework. Section 5 summarizes the main conclusions, limitations, and future research directions. Section 6 lists the references.

2. Related Work

2.1 Deep image colorization

Automatic image colorization has evolved from local optimization and handcrafted propagation rules toward data-driven deep neural networks capable of learning relationships among luminance, semantics, objects, scenes, and plausible colors. A major challenge is that reconstruction losses alone may encourage conservative predictions, particularly in regions for which several colors are plausible. This can result in low saturation or averaged colors.

Recent transformer-based methods have sought to improve long-range reasoning. ColorFormer introduces a color-memory-assisted hybrid-attention transformer that explicitly addresses semantic consistency and color richness [2]. Its global-local attention design is useful because color decisions often depend simultaneously on local details and broad semantic context. UniColor extends this direction by unifying several modes of user guidance, including exemplars, strokes, text, and mixtures of conditions within one transformer-based framework [3]. These methods illustrate the usefulness of flexible semantic representations for ambiguous colorization.

Generative priors provide another strategy. BigColor separates color synthesis from the burden of generating image structure and learns a dedicated generative color prior that supports diverse natural-image colorization [4]. DDColor instead employs two complementary decoders. Its pixel decoder reconstructs spatial detail, while a query-based color decoder learns semantic-aware color queries and interacts with multiscale representations through attention [5]. These approaches are relevant to the present work because they demonstrate two important principles: rich colorization depends on semantic context, and color bleeding is strongly associated with insufficient boundary-aware representation.

Although these image methods can produce visually convincing isolated frames, simply applying them independently to every video frame does not guarantee temporal consistency. Slight changes in motion, cropping, lighting, or feature activation may cause different colors to be generated for the same object across consecutive frames. Therefore, video colorization requires mechanisms that constrain or propagate information across time.

2.2 Exemplar-based and multimodal colorization

Exemplar-based colorization conditions the output on one or more color reference images. The objective is not merely to generate a plausible color image, but to transfer relevant chromatic characteristics from an exemplar while respecting semantic correspondence. This provides greater user control than fully automatic colorization and can be particularly important when historically correct, artistically specified, or identity-specific colors are required.

However, exemplar-based color transfer can fail when the reference and target differ substantially. A semantic feature in the grayscale target may have no accurate match in the exemplar. Conversely, visually similar regions can represent semantically different objects. Errors in correspondence can therefore cause color bleeding, incorrect transfer, incomplete colorization, or local desaturation.

UniColor demonstrates that exemplar guidance can be integrated with other conditioning modes in a common transformer architecture [3]. More recent exemplar video methods have increasingly combined correspondence with temporal reasoning rather than treating each target frame independently.

The NTIRE 2023 Video Colorization Challenge reflects the growing emphasis on both visual quality and color-distribution consistency [6]. Temporal Consistent Automatic Video Colorization uses semantic correspondence to connect a reference image with subsequent frames and combines reference guidance with the previously colorized frame [7]. The method specifically targets the weakness of approaches that maintain only adjacent-frame consistency while failing over longer temporal intervals.

OmniFusion employs OmniMotion and diffusion priors to improve reference-based video colorization under incomplete propagation and correspondence uncertainty [11]. L-C4 uses language conditioning, temporally deformable attention, and cross-clip fusion to address both creative color specification and temporal consistency [14]. These methods show that the field is moving toward increasingly expressive guidance. Nevertheless, an efficient exemplar pipeline remains valuable when reliable reference frames are available and computational simplicity is important.

2.3 Temporal consistency in video colorization

Temporal consistency is one of the defining differences between still-image and video colorization. Even a high-quality frame-wise model can produce unpleasant video because human observers are highly sensitive to repeated color fluctuations over time. Common artifacts include flickering, sudden hue changes, saturation variation, color drift, unstable boundaries, and progressive accumulation of propagation errors.

TCVC addresses this problem by propagating frame-level deep features bidirectionally and applying self-regularization to minimize differences between predictions generated from different temporal contexts [8]. Its design shows that temporal modeling can be incorporated without requiring direct ground-truth color video for all components of temporal self-regularization.

BiSTNet combines semantic exemplar matching with bidirectional temporal feature fusion and semantic boundary modeling [9]. Importantly, recent comparative evidence indicates that using references from both ends of a sequence and employing bidirectional fusion can improve colorization quality and consistency compared with a one-sided exemplar configuration.

Long-range consistency is another concern. Chen et al. proposed a long-term spatiotemporal dependency model combining a parallel convolutional-transformer block, nonlocal operations, and a temporal linkage subnet [10]. Zhang et al. explicitly introduced semantic correspondence to preserve long-range color consistency beyond adjacent frames [7]. These methods are especially relevant when objects undergo irregular motion, substantial appearance variation, partial occlusion, or reappearance after a long temporal gap.

Diffusion-based models offer powerful generative priors but also introduce significant computational costs and can still suffer from cross-frame inconsistencies. LatentColorization conditions latent diffusion on temporal information to improve consistency [12]. VanGogh combines multimodal guidance, optical-flow loss, color injection, and luminance replacement to control color and reduce flicker [15]. Palette-guided video colorization uses a common palette as global chromatic context across multiple video windows, addressing the problem that limited-window diffusion inference can otherwise produce inconsistent long-range colors [16].

The present work differs from these approaches by focusing on three practical sources of instability within an exemplar-based recurrent framework: inappropriate fixed color-transfer temperature, nonoptimal smoothing parameters, and abrupt changes in keyframe influence. The proposed solution does not depend on replacing the entire backbone. Instead, it introduces calibration and fusion mechanisms around the existing color-estimation process.

3. Proposed Method

3.1 Overview

Let a grayscale video be represented as $X = \{x_0, x_1, \dots, x_{M-1}\}$, where x_t denotes the grayscale frame at time t . A subset of frames is selected as keyframes, $K = \{K_1, K_2, \dots, K_J\}$, and each keyframe is available in color, either through manual colorization, expert correction, automatic generation followed by human approval, or an external reference source.

The underlying deep exemplar colorization network receives information including the current grayscale target frame, a selected reference color image, and, when recurrent propagation is used, the previously colorized result. Semantic correspondence is established between target and reference features in deep feature space. Color information from semantically related reference regions is then transferred to the target representation and decoded into a color estimate.

For a current target frame x_t , the general operation can be represented as $\omega_t = C(x_t, r, pt-1; \Theta, T)$, where C denotes the exemplar-based colorization network, r is the reference image, $pt-1$ is a previous colorization result when recurrent input is used, Θ represents fixed network parameters, and T is the normalized temperature coefficient governing the softness or concentration of correspondence-based color estimation.

The proposed system introduces three sequential improvements: temperature calibration, deep exemplar color estimation, optimized weighted least-squares refinement, and bidirectional temporal fusion. The temperature coefficient is calibrated once before full-video colorization. Similarly, the weighted least-squares parameters are estimated from representative reference-keyframe pairs. During full inference, every keyframe interval is processed from both temporal directions, and the two resulting color candidates are fused according to temporal proximity.

This design is computationally attractive because the expensive parameter-selection stages are performed once per video or video group rather than independently for every frame.

3.2 Colorization by optimal global temperature estimation

3.2.1 Motivation

In semantic correspondence-based exemplar colorization, an annealing or temperature parameter controls how strongly the network emphasizes the most similar reference locations compared with alternative correspondences. A sharp distribution may transfer highly specific colors when correspondence is accurate but can also amplify matching errors. A softer distribution can reduce sensitivity to individual incorrect matches but may average incompatible colors and produce desaturation.

Using one manually selected coefficient for every video is therefore problematic. Different videos exhibit different scene types, brightness distributions, motions, semantic compositions, reference-target similarities, and chromatic characteristics. A value that works for one video may be unsuitable for another. The proposed approach estimates one video-specific normalized temperature coefficient before processing the complete sequence.

3.2.2 Representative calibration pairs

Assume that N_T representative scenes are selected from the video. For each scene, a known color keyframe K_i is available. The corresponding grayscale image is denoted by G_i . The known color frame serves both as a reference exemplar and as the desired reconstruction target. The grayscale version is colorized repeatedly while varying the temperature coefficient over a candidate set $T = \{0, \Delta T, 2\Delta T, \dots, 1\}$, where ΔT is the predefined search interval.

For each candidate T , the colorization network produces $Y_i(T) = C(G_i, K_i; T)$. The discrepancy between $Y_i(T)$ and the known color frame K_i is then measured. Following the formulation specified in the proposed method, the optimal coefficient is defined by Eq. (1).

$$T_{opt} = \arg \min_T (1/N_T) \sum_{i=1}^{N_T} \|K_i - Y_i(T)\|_2, \quad T \in [0,1] \quad (1)$$

Here, T_{opt} denotes the temperature coefficient considered most suitable for the complete video, K_i is the i -th known color keyframe, $Y_i(T)$ is its reconstructed colorization under coefficient T , and N_T is the number of

representative reference-colorization pairs. For implementation, the error can be evaluated on the full RGB image or, preferably, only on chrominance components after conversion to CIE Lab. The latter prevents luminance differences from dominating the color-selection objective.

3.2.3 Interpretation

The optimization in Eq. (1) replaces manual parameter selection with a repeatable, objective criterion. Since calibration frames should cover different scene conditions, the selected coefficient represents a compromise that minimizes average reconstruction discrepancy across the video rather than overfitting one isolated frame.

The additional calibration cost is limited because the operation is performed only once before whole-video inference. For long videos containing thousands of frames, the preprocessing overhead is expected to remain small relative to the complete colorization workload. The method can be further generalized in future work from a single global T_{opt} to scene-dependent or continuously adaptive coefficients. However, the global formulation offers greater simplicity and avoids frame-to-frame instability caused by frequent parameter changes.

3.3 Smoothing color-estimation results using an optimized weighted least-squares filter

3.3.1 Motivation

The raw output of exemplar-based color estimation can contain spatial artifacts. Incorrect semantic matching may assign an isolated color to a small local region. In other cases, a correct color appears only in part of an object instead of spreading across the entire semantically coherent surface.

A smoothing filter can alleviate these effects by propagating chromatic information over neighboring pixels. However, ordinary smoothing may blur important boundaries and allow one object's color to leak into another. Therefore, an effective color-refinement stage must satisfy two competing objectives: spreading reliable color information smoothly inside homogeneous or semantically connected regions, and preserving boundaries where luminance or structure indicates a likely object transition. A weighted least-squares filter is suitable for this purpose because it can penalize spatial variation differently according to edge strength.

3.3.2 Edge-aware smoothing formulation

Let c denote a raw chrominance channel estimated by the colorization network and L the corresponding luminance guidance image. The refined chrominance output u can be represented conceptually as the minimizer of the energy function below.

$$E(u) = \sum_p (u_p - c_p)^2 + \sigma \sum_{(p,q) \in N} w_{pq}(\lambda) (u_p - u_q)^2$$

Here, p and q denote neighboring pixels, N is the neighborhood system, σ controls spatial smoothing strength, and $w_{pq}(\lambda)$ reduces smoothing across strong boundaries. One possible edge weight is $w_{pq}(\lambda) = 1 / (|L_p - L_q|^\lambda + \epsilon)$, where $\epsilon > 0$ avoids division by zero. Large luminance differences reduce the coupling between neighboring pixels, helping preserve object boundaries. In the terminology of this work, σ is the color-deviation or chromatic-spreading parameter and λ is the boundary-preservation parameter. The candidate intervals are $\sigma \in [100, 500]$ and $\lambda \in [1, 5]$.

3.3.3 Automatic parameter estimation

As with temperature calibration, representative known color keyframes are used. First, their grayscale versions are colorized using the exemplar-based network. Before the final smoothing operation, the raw color estimates are retained. For every candidate pair (σ, λ) , the WLS filter generates a refined result $\hat{Y}_i(\sigma, \lambda) = S(Y_i; \sigma, \lambda)$, where S denotes the edge-preserving weighted least-squares operator.

The optimal pair is selected by minimizing the average reconstruction discrepancy as follows.

$$[\sigma_{opt}, \lambda_{opt}] = \arg \min_{(\sigma, \lambda)} (1/N_S) \sum_{i=1}^{N_S} \|K_i - \hat{Y}_i(\sigma, \lambda)\|_2, \quad \sigma \in [100, 500], \lambda \in [1, 5] \quad (2)$$

Here, K_i is the known color keyframe, $\hat{Y}_i$ is the filtered inferred result, and N_S is the number of reference-keyframe pairs used for smoothing calibration.

3.3.4 Expected effect of parameter optimization

A fixed smoothing strength may produce very different results depending on scene content. For a smooth sky, wall, or garment, stronger chromatic propagation may be desirable. For faces, foliage, thin structures, text, or object intersections, excessive propagation can cause serious boundary contamination.

The optimization in Eq. (2) therefore seeks a balanced video-level parameter pair rather than relying on subjective manual tuning. The anticipated effects include fewer isolated color stains, more complete chromatic coverage of semantically uniform surfaces, reduced leakage across high-contrast boundaries, improved agreement with reference color distributions, and greater repeatability across observers. The optimization is again performed only once before complete-video inference, so its computational contribution is amortized over the full video.

3.4 Between-frame colorization with bidirectional temporal fusion

3.4.1 Limitations of the original keyframe-centered method

Consider three consecutive color keyframes K1, K2, and K3. In a conventional keyframe-centered arrangement, each keyframe serves as the exclusive reference over approximately half of the interval on each side. Thus, in the interval between K1 and K2, frames near K1 are colorized using K1, and after the temporal midpoint the reference abruptly changes to K2. The reference influence can be visualized as a sequence of nonoverlapping triangular regions centered on individual keyframes.

The main problem is that the two keyframes may produce different estimates for the same intermediate frame because of changes in appearance, semantic correspondence, illumination, saturation, or transferred chromatic statistics. A hard reference switch therefore creates a temporal discontinuity even when both estimates are individually plausible.

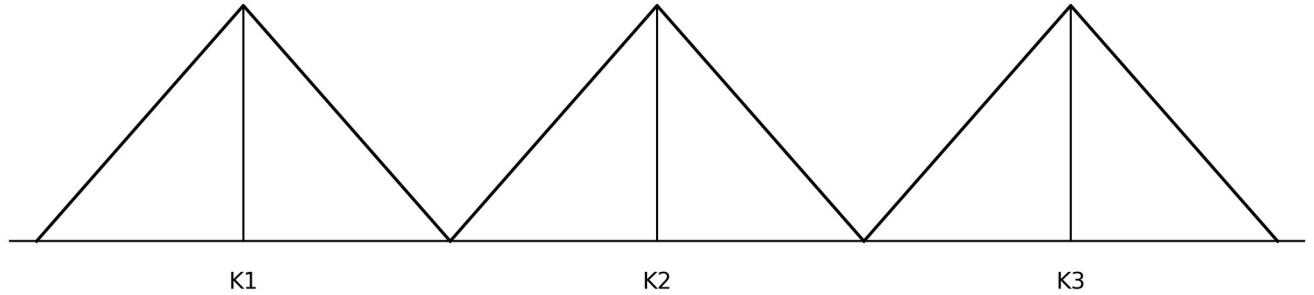


Figure 1. Original between-frame colorization scheme. Each keyframe acts as the reference for approximately half of each neighboring interval, which can create a visible discontinuity near the midpoint.

3.4.2 Proposed overlapping bidirectional scheme

The proposed method extends the influence of each keyframe across the complete interval to the neighboring keyframe. For every intermediate frame, two independent estimates are generated: ωt^f from the earlier, or forward, keyframe, and ωt^b from the later, or backward, keyframe. The two influence regions therefore overlap throughout the complete keyframe interval.

Let an interval consist of $\{x_0, x_1, \dots, x_{N-1}\}$, where x_0 corresponds to the first color reference r_f and x_{N-1} corresponds to the second color reference r_b . The forward and backward temporal distances are $dt^f = t$ and $dt^b = N - 1 - t$, respectively. Their normalized forms are $d\sim t^f = t/(N-1)$ and $d\sim t^b = (N-1-t)/(N-1)$.

Because the weight of a reference should increase as the target becomes closer to it, the complementary proximity weights are $\alpha t^f = (N-1-t)/(N-1)$ and $\alpha t^b = t/(N-1)$. These satisfy $\alpha t^f + \alpha t^b = 1$. The fused chromatic estimate is therefore defined by Eq. (3).

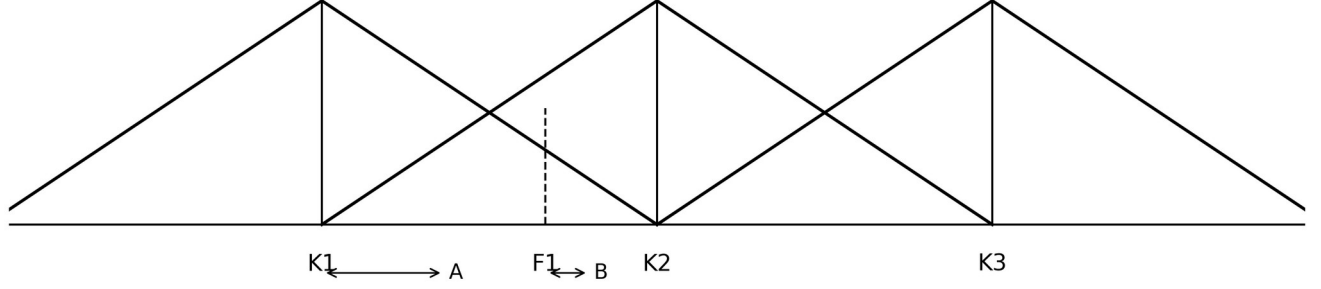


Figure 2. Proposed bidirectional between-frame colorization scheme. Each intermediate frame is influenced by both neighboring keyframes, with larger weight assigned to the temporally closer reference.

$$p_t = ((N - 1 - t)/(N - 1)) \omega_t^f + (t/(N - 1)) \omega_t^b \quad (3)$$

At the first frame, $p_0 = \omega_0^f$ because the forward weight is 1 and the backward weight is 0. At the last frame, $p_{N-1} = \omega_{N-1}^b$. At the temporal midpoint, both estimates receive approximately equal influence. This ensures that the first and last frames are not contaminated by the opposite endpoint, while intermediate frames transition gradually from one reference to the other.

Suppose an intermediate frame F_1 lies closer to K_1 than to K_2 . The colorization obtained from K_1 receives the larger coefficient B , while the result obtained from K_2 receives the smaller coefficient A , with $B > A$. As the sequence advances toward K_2 , B decreases continuously and A increases. There is therefore no hard midpoint switch. Local correspondence errors from one reference can be moderated by the estimate from the opposite temporal direction, color transitions become continuous functions of temporal position, and endpoint colors remain exactly controlled by their corresponding reference frames.

This principle is consistent with broader evidence that bidirectional temporal information improves exemplar-based video colorization and that using two temporally separated references can outperform a one-sided reference configuration [9].

3.4.3 Complete inference procedure

For every pair of consecutive keyframes, the complete inference procedure is:

- Perform forward exemplar colorization from the earlier keyframe.
- Perform backward exemplar colorization from the later keyframe.
- Apply the optimized temperature coefficient T_{opt} .
- Refine estimated chromatic channels using $(\sigma_{\text{opt}}, \lambda_{\text{opt}})$.
- Fuse the two estimates using Eq. (3).
- Combine the fused chrominance with the original target luminance.

Accordingly, the final output sequence is $P = \{p_0, p_1, \dots, p_{M-1}\}$.

4. Experimental Results

4.1 Implementation details

The proposed framework should be evaluated using a standard deep exemplar-based video-colorization backbone capable of semantic correspondence and recurrent color propagation. The architecture itself is kept fixed during ablation so that performance differences can be attributed specifically to the three proposed components.

The experimental configurations should contain at least four variants: the baseline original exemplar model with fixed parameters and hard reference switching; Variant A, which adds T_{opt} ; Variant B, which adds both T_{opt} and

optimized WLS refinement; and the full proposed model, which adds bidirectional temporal fusion. This incremental ablation design allows the effect of each component to be measured independently.

For calibration, representative keyframes should be selected from semantically diverse parts of the test video. A useful selection policy should cover differences in indoor and outdoor scenes, illumination, motion intensity, dominant colors, object categories, and reference-target similarity. To avoid evaluation leakage, the final test frames used for reported benchmark results should not be reused to select hyperparameters where the protocol requires a strict training-validation-test separation.

Suitable evaluation datasets include DAVIS, Videvo, and the NTIRE Video Colorization Challenge data. The Videvo evaluation set has been used extensively in prior temporal-colorization studies, while the NTIRE challenge provides an established benchmark emphasizing both perceptual quality and color-distribution consistency [6,13].

The implementation should maintain identical resizing, preprocessing, reference selection, and backbone checkpoint conditions across all ablation variants. Where CIELab is used, the grayscale L channel should remain unchanged while the network predicts or transfers chrominance information. The temperature search should evaluate $T \in [0,1]$ at a predefined regular interval. The WLS parameter search should cover $\sigma \in [100,500]$ and $\lambda \in [1,5]$. The final submitted version should report the exact search step sizes for reproducibility.

4.2 Evaluation metrics

No single metric is sufficient for video colorization because the task combines reconstruction quality, perceptual realism, chromatic plausibility, and temporal stability. The following evaluation dimensions are recommended.

4.2.1 PSNR and SSIM

Peak signal-to-noise ratio and structural similarity measure reconstruction fidelity against a known color ground truth. They are useful when the source grayscale video is derived directly from an existing color video.

4.2.2 LPIPS and FID

LPIPS measures perceptual feature-space difference, whereas Fréchet Inception Distance evaluates distribution-level similarity between generated and real images. These measures are useful because pixel-level errors can penalize a plausible alternative color even when it is visually convincing.

4.2.3 Color Distribution Consistency

Color Distribution Consistency (CDC) measures stability of color distributions across the video and has been used prominently in the NTIRE 2023 Video Colorization Challenge [6]. A lower CDC value indicates more stable chromatic distribution over time.

4.2.4 Temporal warping error

Optical-flow-based warping error can compare a colorized frame with a motion-compensated adjacent output. Large disagreement indicates instability beyond what is expected from actual scene motion. Recent surveys show that modern video-colorization evaluations frequently combine PSNR, SSIM, LPIPS, FID, CDC, and warping-based temporal criteria rather than relying on only one measure [13].

4.3 Evaluation of video colorization

4.3.1 Performance and analysis with initial settings

The original exemplar-based configuration uses fixed color-estimation and smoothing parameters. Each keyframe controls only part of the neighboring temporal interval, producing an abrupt reference change near the midpoint.

Three characteristic weaknesses are expected. The first is scene-dependent color estimation. A fixed temperature can produce sufficiently vivid and accurate colors in some scenes but overly diffuse or incorrectly concentrated correspondence in others. Bright scenes with simple semantics may require a different correspondence sharpness from crowded scenes containing multiple objects and complex textures.

The second weakness is inconsistent spatial color spread. Under insufficient smoothing, a semantically uniform object may be only partially colorized or display localized color noise. Under excessive smoothing, chromatic information can cross high-contrast boundaries and contaminate neighboring objects.

The third weakness is temporal blinking. When the active exemplar changes abruptly from one keyframe to another, a visible jump can occur even though consecutive grayscale frames remain structurally similar. The effect is particularly noticeable when corresponding keyframes contain different saturation, lighting, or imperfectly matched colors. These observations motivate the subsequent components.

4.3.2 Performance and analysis with optimal temperature estimation

With T_{opt} , the correspondence sharpness is no longer chosen only by visual judgment. Instead, representative color-reference pairs provide an objective calibration criterion. The anticipated improvement should be examined in three situations: close target-exemplar correspondence, moderate semantic mismatch, and substantial scene diversity across the full sequence.

For a frame whose content closely matches the exemplar, the optimized coefficient should preserve confident chromatic transfer without unnecessarily averaging unrelated reference locations. Under moderate semantic mismatch, the optimized temperature should prevent a single unreliable correspondence from dominating the output. Across multiple scenes, the globally selected coefficient should provide more stable average performance than an arbitrarily chosen fixed value.

The primary quantitative evidence should be the average change in PSNR, SSIM, LPIPS, or chromatic reconstruction error relative to the baseline. A temperature-sensitivity plot should also report the objective in Eq. (1) as a function of T . Such a curve would demonstrate whether the optimization has a clear minimum and how sensitive the video is to parameter selection.

After measurements are obtained, an appropriate interpretation is that lower values may produce excessive averaging and reduced saturation, whereas overly large values can increase sensitivity to mismatched semantic regions. The optimized coefficient should achieve the best average balance across the selected calibration scenes. The exact numerical value of T_{opt} must be inserted after execution of the experiment.

4.3.3 Colorization performance and analysis with the weighted least-squares filter

The WLS refinement stage should be evaluated through both full-reference metrics and close visual inspection of challenging spatial structures. Particular attention should be paid to object-background boundaries, faces and skin regions, clothing, foliage, sky-object borders, thin structures, regions containing repeated texture, and low-contrast neighboring objects.

Compared with unoptimized filtering, the optimal pair $(\sigma_{\text{opt}}, \lambda_{\text{opt}})$ is expected to provide more uniform chromatic coverage inside objects while retaining stronger separation at significant luminance boundaries.

The behavior of the two parameters should be analyzed separately. Increasing the color-spreading strength too far may reduce local noise but can introduce oversmoothing. Insufficient spreading can preserve texture yet leave patchy or incomplete color. Similarly, weak boundary protection may cause color leakage, while excessively strong boundary protection can prevent useful propagation inside objects with internal texture.

A two-dimensional error surface over (σ, λ) is therefore recommended. It would provide stronger scientific evidence than reporting only the final optimum because it demonstrates the robustness and interaction of the two parameters. After measurements are obtained, the analysis should discuss whether optimized WLS reduces isolated chromatic artifacts, improves within-object color homogeneity, and maintains stronger discontinuities across object boundaries.

4.3.4 Performance and analysis of bidirectional temporal fusion

The main objective of bidirectional temporal fusion is to eliminate discontinuity at the reference-switching point between two keyframes. The baseline method can be tested by plotting average chromatic difference between consecutive output frames, $D_t = (1/|\Omega|) \sum_{p \in \Omega} \|C_t(p) - C_{t-1}(p)\|_2$, where C_t denotes the chrominance of the output at frame t , p is a pixel location, and Ω is the image domain.

Under a hard keyframe switch, D_t is expected to show an abnormal peak near the midpoint if the two reference-conditioned predictions differ. Under the proposed fusion in Eq. (3), this peak should be substantially reduced because reference contributions vary continuously.

The qualitative analysis should compare consecutive frames surrounding the midpoint of each interval. The strongest evidence would include side-by-side results for the output using only K1, the output using only K2, the baseline hard-switch result, and the proposed weighted fusion. The fused output should preserve stronger influence from the temporally closer keyframe. This behavior is conceptually consistent with recent bidirectional exemplar methods such as BiSTNet, which explicitly combine information from references at opposite temporal directions to improve color propagation [9].

The proposed method should be particularly effective when the two keyframes are reasonably similar but not chromatically identical. Under extremely large scene changes, however, direct linear temporal fusion may blend incompatible content. This limitation is discussed in Section 5.

4.4 Ablation analysis

A complete ablation table should be reported in the final manuscript. The structure below associates each component with the quality dimensions it is designed to improve. The entries marked 'Measured' must be replaced with real values after experiments are executed.

Method	Optimal T	Optimized WLS	Bidirectional fusion	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	CDC $\downarrow$	Warp error $\downarrow$
Baseline	No	No	No	Measured	Measured	Measured	Measured	Measured	Measured
Variant A	Yes	No	No	Measured	Measured	Measured	Measured	Measured	Measured
Variant B	Yes	Yes	No	Measured	Measured	Measured	Measured	Measured	Measured
Full proposed method	Yes	Yes	Yes	Measured	Measured	Measured	Measured	Measured	Measured

Table 1. Recommended ablation structure for the baseline and three incremental variants. Higher is better for PSNR and SSIM; lower is better for LPIPS, FID, CDC, and temporal warping error.

The expected pattern is not necessarily that every component improves every metric equally. Temperature optimization primarily targets reference-color agreement. WLS optimization primarily improves spatial chromatic coherence and boundaries. Bidirectional fusion primarily improves temporal stability. Consequently, the strongest scientific evaluation should associate each mechanism with the metric it is explicitly designed to optimize.

4.5 Comparison with recent work

Recent approaches provide useful comparative categories. TCVC focuses on bidirectional deep-feature propagation and self-regularization [8]. BiSTNet combines semantic exemplars with bidirectional temporal fusion and semantic boundary priors [9]. Long-term spatiotemporal dependency modeling incorporates transformer and linkage operations for irregular motion [10]. OmniFusion combines motion modeling and diffusion priors [11]. LatentColorization applies latent diffusion with temporal conditioning [12]. VanGogh and palette-guided methods represent newer diffusion-based directions with multimodal, optical-flow, and global color context [15,16].

The proposed method should not be positioned as a direct architectural replacement for every one of these systems. Its principal contribution is a practical enhancement of deep exemplar-based colorization through video-specific parameter calibration and simple bidirectional distance-based fusion.

Its potential advantages include objective rather than purely manual temperature selection; objective joint selection of spatial smoothing and boundary-preservation parameters; smooth transition between successive exemplar keyframes; compatibility with an existing recurrent exemplar-colorization backbone; no requirement for a large new diffusion model; and limited additional calibration overhead relative to full-video inference. However, direct state-of-the-art claims should be made only after actual benchmark measurements are completed.

5. Conclusion

This paper presented a deep exemplar-based video-colorization framework designed to improve chromatic accuracy, spatial smoothness, boundary preservation, and temporal consistency. The method addresses three practical weaknesses of conventional exemplar-based processing: the use of a fixed temperature coefficient for semantically diverse videos, manually selected spatial-smoothing parameters, and abrupt reference changes between neighboring keyframes.

First, an optimal normalized temperature coefficient is estimated before whole-video colorization. Representative color keyframes are converted to grayscale, recolorized under different candidate temperatures, and compared against their known color counterparts. The coefficient minimizing average reconstruction discrepancy is selected as T_{opt} .

Second, weighted least-squares filtering is optimized using representative keyframe pairs. The proposed procedure searches for the combination of chromatic-spreading and boundary-preservation parameters that minimizes reconstruction error. This is intended to suppress isolated color artifacts, improve spatial coverage, and reduce color leakage across semantic boundaries.

Third, a bidirectional temporal fusion scheme replaces abrupt midpoint reference switching. Every frame between neighboring keyframes is colorized from both temporal directions, and the two predictions are blended using complementary normalized weights based on temporal distance. The first reference dominates near the start of an interval, the second reference dominates near the end, and their influences vary continuously in between. This strategy is designed to suppress blinking and produce smoother color transitions.

The method has several limitations. A single globally optimized temperature may still be insufficient for long videos containing radically different scenes. The same limitation applies to one global WLS parameter pair. Future research should therefore investigate scene-adaptive or continuously predicted parameters.

The linear temporal fusion model also assumes that both endpoint-conditioned predictions remain meaningful throughout the interval. This assumption may fail under hard scene cuts, severe occlusion, rapid object appearance or disappearance, or major semantic changes. Future extensions could incorporate scene-cut detection, confidence-aware weighting, optical-flow reliability, semantic-match confidence, learned fusion networks, or occlusion masks.

Another promising direction is integration with recent transformer and diffusion models. Modern video-colorization systems increasingly use long-range attention, language guidance, generative priors, optical flow, global palettes, and multimodal constraints [14-16]. The proposed optimization and temporal weighting principles could potentially be incorporated into these architectures as calibration or post-processing modules.

Most importantly, the complete method should be validated quantitatively on established datasets using reconstruction, perceptual, color-distribution, and temporal metrics. Once measured experimental results are available, a comprehensive ablation study should determine the individual contribution of T_{opt} , optimized WLS refinement, and bidirectional temporal fusion.

Overall, the proposed framework provides a practical approach to improving deep exemplar-based video colorization without requiring a complete redesign of the underlying network. By combining objective parameter estimation, edge-aware chromatic refinement, and continuous bidirectional keyframe fusion, the method directly targets three major sources of degradation in reference-guided video colorization.

6. References

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Draft Status Note

The numerical experimental values for PSNR, SSIM, LPIPS, FID, CDC, temporal warping error, T_{opt} , σ_{opt} , and λ_{opt} remain to be inserted after actual experiments are run. No numerical performance claim has been fabricated in this manuscript draft. The experimental section is written as a scientifically defensible evaluation plan and qualitative analysis framework until measured results become available.